

New Insights into the Propulsion Power Prediction of Cruise Ships





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Context



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Why Energy Efficiency

- Shipping is ~2.8% of global carbon emissions – *International Maritime Organization* [1]
- Reduce shipping carbon intensity by 40% by 2030 and 70% intensity by 2050

Luxury Cruise Ships

- Floating city (and also a hotel...)
- Consumption: **3-4 GWh ~ hundreds of tons of fuel per week**
- Propulsion accounts for between **40-60%** of consumption
- Increasing amounts of sensor data being collected



Propulsion Efficiency

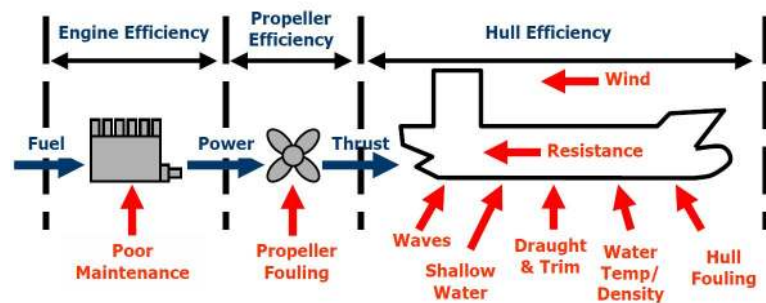


Fig 1: Variables effecting propulsion power. Adapted from Pedersen B. P. (2009) Prediction of full scale propulsion power using artificial neural networks



International Maritime Organization:

- Shipping is ~2.8% of global carbon emissions – International Maritime Organization [1]
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Main Challenges:

- Performance evaluation over a given period (voyage, month, year)
 - Performance degradation over time
 - **Prediction over given route/conditions**
- Computational Fluid Dynamics (CFD)
 - Resistive effects: wind, wave, trim and draft, water depth, etc.

Solution: Use sensor and weather data and Machine Learning to predict propulsion power given operational and external conditions

Literature Review and Gaps

Literature Review

- Recent rise in propulsion power prediction papers using ML [3-7]
- Most papers apply aggregation (1-10 minute windows), treat each window as an independent observation
- Approaches include kNN, Random Forest, Multilayer Perceptron [4, 5], ML regression [7], Gaussian Processes [6] ...
- Standard approach in the literature:
 - Apply filters
 - Extract features
 - Aggregate data
 - **Shuffle and split into train/validation/test**

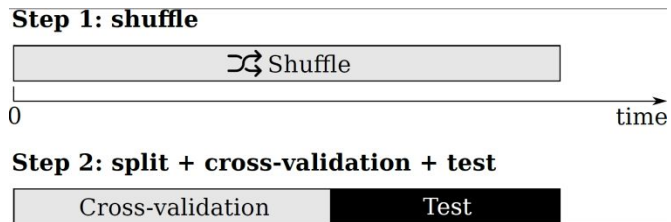


Fig 2: Shuffle Split Approach

Methodological issues with current approaches:

- Cross-talk problem: Separation of high frequency data into train/validation sets **after shuffling** causes very close analogs to the validation data to appear in the training set (Fig. 3), while future, unseen data may not relate as closely to the training data.
- Overlapping windows: Using rolling averages of data exacerbates the cross-talk problem. The validation accuracy obtained from model training may be overestimated due to strong overfitting
- Parsimony: Using the current approach may lead to larger/more complex models than needed for

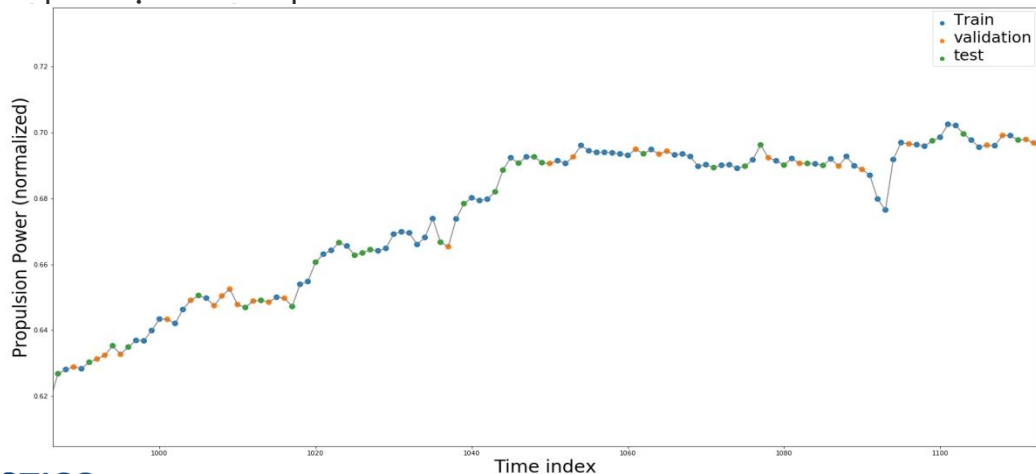


Fig 3: Shuffled data

Research Questions

1. Given a propulsion power prediction task, what is the most appropriate splitting approach (train/validation/test) to obtain a model leading to similar validation and test errors (i.e., providing more confidence for implementation in operational conditions)?
2. Given a fixed size of a model, which splitting approach gives the minimum test error (i.e., which approach generalizes better)?
3. Given a specific splitting approach, does a small model perform as well on a future test as a large model (i.e. principle of parsimony)?

Methodology and Experimentation



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Shuffle-split approach

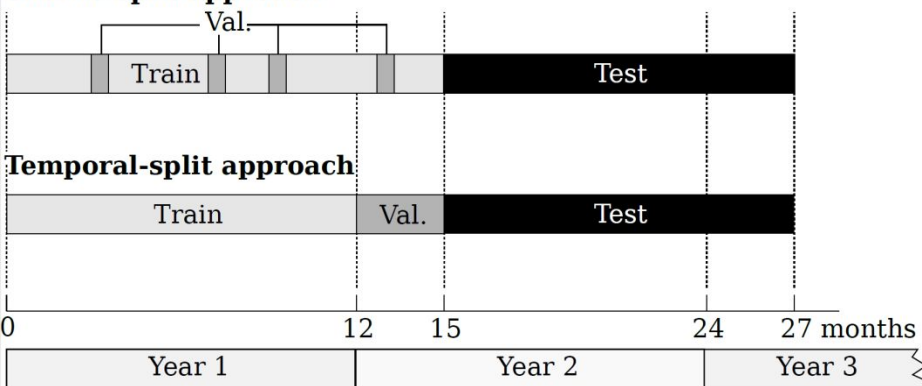


Fig 4: Shuffle-split vs Temporal-split approaches

Parameter	Value
Batch Size	128
Optimizer	Adam [17]
Learning Rate	5e-4
Max Epochs	50
Loss	Mean Absolute Percentage Error (MAPE)

TABLE I
SHARED HYPERPARAMETERS

Parameter	Possible values
Layers	10, 15, 20
Nodes per layer	50, 100, 200, 300, 400

TABLE II

Hyperparameter	Possible values
Layers	1, 3, 5, 10
Nodes per layer	5, 10, 15, 20, 30, 40, 50

TABLE III

HYPERPARAMETERS GRID FOR THE LARGE MODEL HYPERPARAMETERS GRID FOR THE SMALL MODEL

Global Framework

- Methodological Framework:
 - Shuffle-split approach
 - Temporal-split approach
- Model Dimension Analysis:
 - MLP architecture
 - Grid-search and cross-validation for hyper-parameter selection
 - Cross-comparison
- Realistic evaluation:
 - Test each model on future, unseen data
- Dataset:
 - 27 months of sensor and weather data
 - First 15 months for training/validation and next 12 months for testing

Shuffle-split vs Temporal split

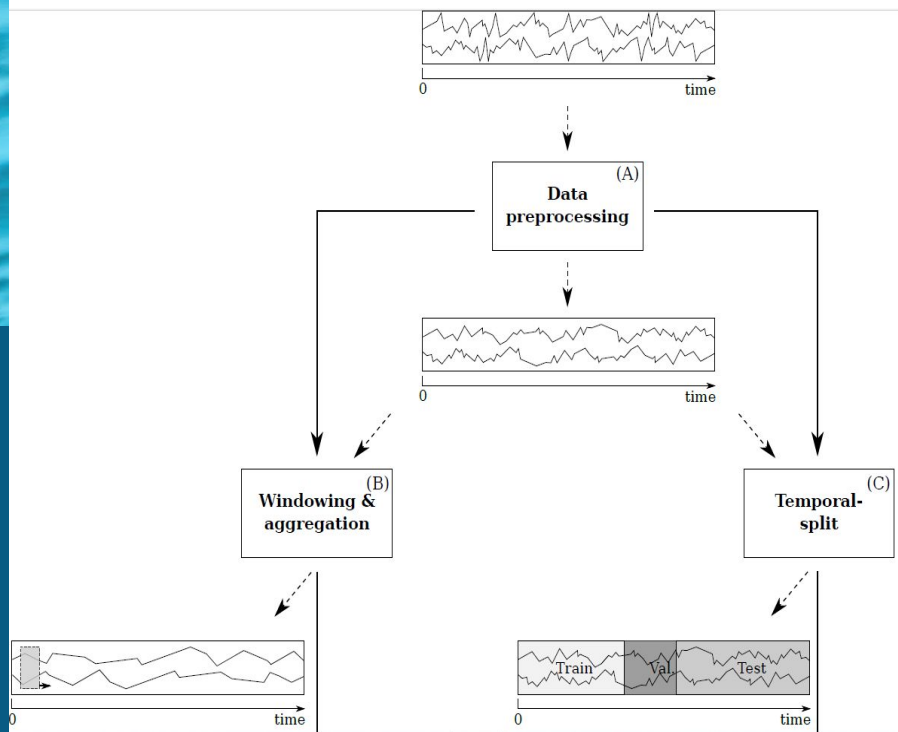
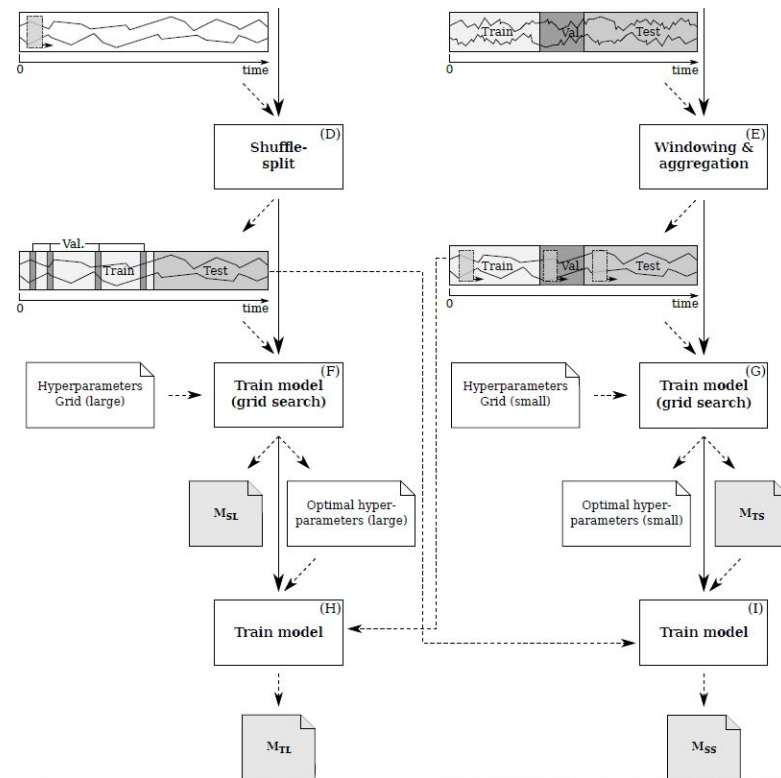


Fig 5: Global Methodological Framework



M_{SL} : Model trained with shuffle – split data, large hyperparameter values
 M_{TS} : Model trained with temporal – split data, small hyperparameter values
 M_{TL} : Model trained with temporal – split data, large hyperparameter values
 M_{SS} : Model trained with shuffle – split data, small hyperparameter values

Results



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Results

- Grid-search results using shuffle-split and temporal-split strategies

Model	Layers	Nodes per layer
M_{SL}	20	400
M_{TS}	3	30

TABLE IV

GRID SEARCH RESULTS: OPTIMAL HYPERPARAMETERS FOUND DURING STEPS (F) AND (G) IN FIGURE 3

As expected, shuffle-split favored higher values of the grid compared to temporal-split

- Validation and test losses:

Model	Val. Loss	Test Loss	Difference
M_{SL}	0.75	6.33	88.3%
M_{TL}	4.45	5.41	17.7%
M_{SS}	2.75	5.52	50%
M_{TS}	4.35	5.14	15.2%

TABLE V

COMPARISON OF VALIDATION AND TEST LOSSES FOR THE FOUR MODELS

¹Models are identified as M_{xy} , where $x \in \{S, T\}$ denotes the use of the Shuffle- or Temporal-split dataset, and $y \in \{S, L\}$ denotes the use of hyperparameters coming from the Small or Large grid for training.

Results

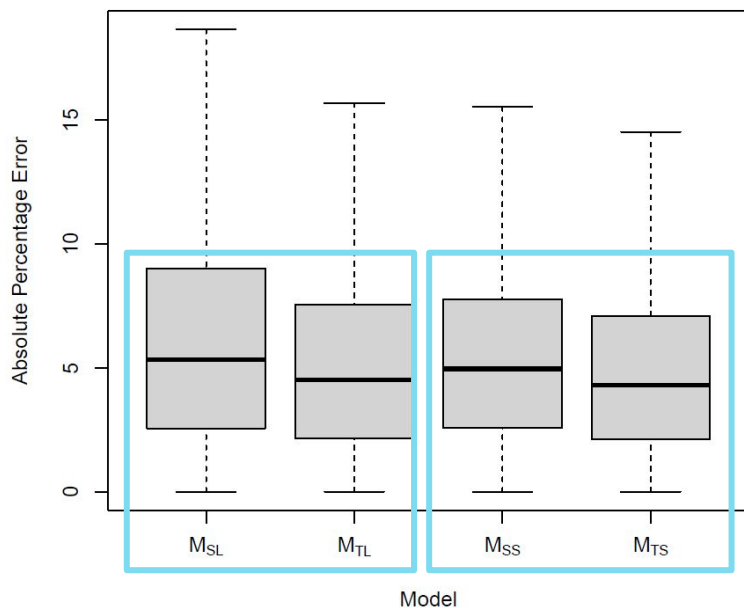


Fig 5: Boxplot distributions of absolute percentage errors of all models

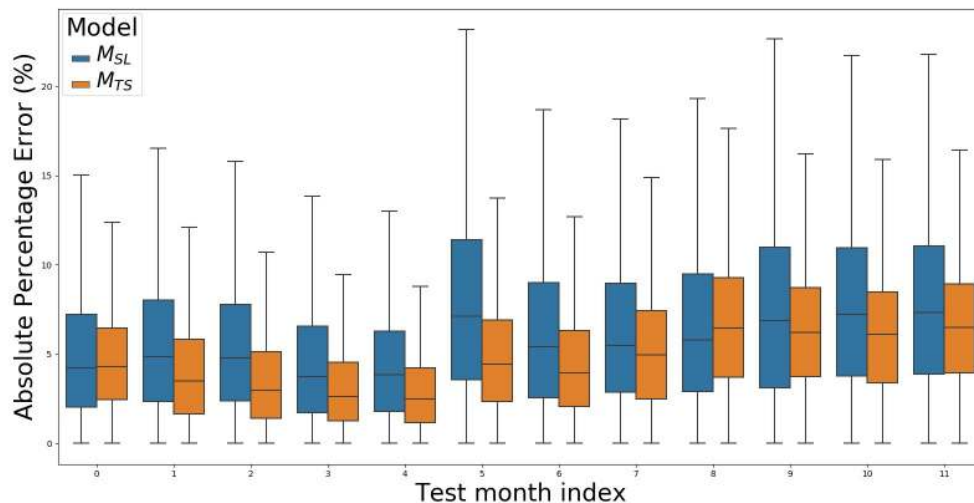


Fig 6: Absolute percentage error vs time

- Given a fixed model size, the temporal split approach always performs better on future test data (blue boxes in Fig. 5)
- Given a fixed data selection strategy, smaller architectures, smaller architectures are slightly better than larger ones
- Temporally-split model performs consistently better over time (Fig. 6)

Conclusions and Perspectives

Conclusions and Perspectives

- Shuffle-split approach leads to a cross talk problem, hence overestimation of model accuracy during training/evaluation phase
- Temporally splitting the training/validation data can reduce/eliminate this problem
- Smaller models may be sufficient to explain the variance in the data
- Reanalysis vs prediction: shuffle-split methodology could be used to analyze historical data and sensitivity to certain input variables, but not necessarily for prediction purposes
- Perspective work:
 - Other splitting strategies: eg. by trips
 - Time-series based approaches
 - Unsupervised models: eg. clustering based on power consumption or deviation from theoretical value, trip-based clustering, etc.

Selected References

1. I. M. Organization, "Third IMO GHG Study 2014," <https://www.imo.org/en/OurWork/Environment/Pages/Greenhouse-Gas-Studies-2014>.
2. Holtrop, J.. (1984). A statistical re-analysis of resistance and propulsion data. *International Shipbuilding Progress*. 31. 272-276.
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4. Pavlos Karagiannidis, & Nikos Themelis (2021). Data-driven modelling of ship propulsion and the effect of data pre-processing on the prediction of ship fuel consumption and speed loss. *Ocean Engineering*, 222.
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Thanks!

